

Article

Analysis of the Spatial–Temporal Distribution Characteristics of NO₂ and Their Influencing Factors in the Yangtze River Delta Based on Sentinel-5P Satellite Data

Xiaohui Guo ^{1,2,3}, Zhen Zhang ^{1,2,3,*} , Zongcai Cai ^{1,2,3}, Leilei Wang ^{1,2,3}, Zhengnan Gu ^{1,2,3}, Yangyang Xu ^{1,2,3} and Jinbiao Zhao ^{1,2,3}

¹ School of Geomatics, Anhui University of Science and Technology, Huainan 232001, China

² Key Laboratory of Aviation-Aerospace-Ground Cooperative Monitoring and Early Warning of Coal Mining-Induced Disasters of Anhui Higher Education Institutes, Anhui University of Science and Technology, Huainan 232001, China

³ Coal Industry Engineering Research Center of Mining Area Environmental and Disaster Cooperative Monitoring, Anhui University of Science and Technology, Huainan 232001, China

* Correspondence: zhangzhen@aust.edu.cn

Abstract: The recent rapid economic development in the Yangtze River Delta (YRD) has led to atmospheric destruction; therefore, it is imperative to solve the issue of atmospheric environmental pollution to ensure stable long-term development. Based on the NO₂ column concentration observed by the TROPOMI (a tropospheric monitoring instrument) on the Sentinel-5P, the spatial–temporal distribution characteristics of the NO₂ column concentration in the YRD from 2019 to 2020 were analyzed using the Google Earth Engine (GEE) platform, and the Geographical Detector (Geodetector) model was used to determine the driving factors of the NO₂ column concentration. The results show that the correlation between the NO₂ column concentration and the ground-monitored NO₂ concentrations reached 70%. The annual variation trend of the NO₂ column concentration exhibited a ‘U’-shaped curve, with the characteristics of ‘high in winter and low in summer, with a transition between spring and autumn’. It exhibited obvious agglomeration characteristics in terms of the spatial distribution, with a high-value agglomeration in the central region of the YRD, followed by the northern region, and a low-value agglomeration in the southern region, with higher altitudes. The change in the NO₂ column concentration in the YRD was affected by both physical geographical factors and socio-economic factors; it is clear that the influence of socio-economic factors has increased.

Keywords: Sentinel-5P; NO₂ column concentration; Yangtze River Delta; spatial–temporal distribution characteristics; geographical detector model



Citation: Guo, X.; Zhang, Z.; Cai, Z.; Wang, L.; Gu, Z.; Xu, Y.; Zhao, J. Analysis of the Spatial–Temporal Distribution Characteristics of NO₂ and Their Influencing Factors in the Yangtze River Delta Based on Sentinel-5P Satellite Data. *Atmosphere* **2022**, *13*, 1923. <https://doi.org/10.3390/atmos13111923>

Academic Editor: David F. Plusquellic

Received: 28 August 2022

Accepted: 13 November 2022

Published: 18 November 2022

Publisher’s Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Copyright: © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

In recent years, with rapid developments in urbanization and industrialization, China’s economy has continued to grow; however, this economic growth has led to the burning of a large amount of fossil fuels, resulting in a substantial increase in the concentrations of nitrogen dioxide (NO₂) [1]. NO₂ is a major tropospheric air pollutant, and is involved in the formation of ozone (O₃), acid rain, and aerosol particles, which affect the radiant intensity of the climate system [2]. In addition, high concentrations of NO₂ on the ground can lead to cardiovascular and respiratory diseases [3]; this also affects the growth of vegetation, leading to the destruction of the local ecological environment [4]. Therefore, it is of great significance to analyze the spatial–temporal distribution characteristics of NO₂ and their driving factors for air-pollution control and improving air quality.

As ground-monitoring technology continues to develop, the spatial–temporal resolution of data is also improving, and the number of monitoring sites continues to increase; however, it is still difficult to obtain NO₂ concentration data with widespread geographical

coverage and high temporal continuity. Therefore, remote sensing technology has become crucial for effectively monitoring atmospheric trace gases, owing to its high spatial–temporal resolution. Since 1995, a series of satellite sensors for observing atmospheric NO₂ has been developed. For example, the ERS-2 satellite carrying the Global Ozone Monitoring Experiment (GOME) sensor, was an early satellite used to monitor NO₂; however, its spatial resolution is low, at 320 km × 40 km [5,6]. In 2002, the Scanning Imaging Absorption Spectrometer for Atmospheric Cartography (SCIAMACHY) sensor, carried by the ENVISAT satellite, was launched, with an improved spatial resolution [7]. In 2004, the Ozone Monitoring Instrument (OMI) sensor, carried by the AURA satellite, was launched, which further increased the spatial resolution to 24 km × 13 km, which greatly benefited research on atmospheric NO₂ [8,9]. Through more in-depth research, scholars have proposed higher requirements for the temporal and spatial resolution of satellite remote-sensing data. Furthermore, to bridge the data gap between SCIAMACHY and the upcoming Sentinel-5, the Sentinel-5P satellite equipped with the tropospheric monitoring instrument (TROPOMI) sensor was launched in 2017 [10], when it began its task of global trace gas monitoring. The TROPOMI sensor is an air pollution monitoring sensor that has the highest spatial resolution thus far, at 3.5 km × 7 km [11].

At present, the spatial–temporal variation of the tropospheric NO₂ column concentration observed by the Sentinel-5P satellite is primarily analyzed using different administrative divisions. Some researchers have explored the spatial–temporal variation of regional NO₂ column concentration at the national, provincial, and city scales, verifying the feasibility of exploring the spatial–temporal variation of the NO₂ column concentration through remote sensing [12–18]. Further, urban agglomeration is a major factor in promoting regional, social, and economic development [19]. In the process of traditional extensive urbanization, air pollution problems inevitably become prevalent, which are usually caused by excessive population agglomeration, traffic congestion, and excessive energy consumption [20]. In particular, the economically developed eastern urban agglomeration in China is a severely polluted area. In 2019, the ‘Outline of the Yangtze River Delta Regional Integrated Development Plan’ issued by the Central Committee of the Communist Party of China (CPC) and the State Council, proposed the goal of significantly improving the eco-environmental protection ability and joint management in the Yangtze River Delta (YRD) [21]. Research on the spatial–temporal variation of NO₂ in the YRD is still in its nascency [22]. Therefore, to achieve the dual goals of high-quality urbanization development and ecological civilization construction, it is imperative to understand the spatial–temporal evolution of the NO₂ column concentration in the YRD, clarify its influencing factors, and propose targeted prevention and control measures.

In addition, when studying spatial–temporal changes, many researchers have begun to focus on the factors affecting NO₂. At present, most research on these influencing factors uses the linear regression method. For example, Zheng et al. [17] used ordinary least square (OLS) and geographically weighted regression (GWR) to analyze the spatial distribution of NO₂ in the Guangdong–Hong Kong–Macao Greater Bay Area, demonstrating that the intensity of human activities, vegetation, and elevation were significantly correlated with the NO₂ column concentration. Li et al. [23] used spatial measurement models, such as the spatial lag model (SLM) and spatial error model (SEM) to analyze the characteristics of air pollution in China, finding that the temperature and wind speed are the most important meteorological factors affecting air pollution. These methods can represent linear relationships but cannot capture non-linear relationships. In fact, there is no strict statistical standard governing linear relationships between environmental pollution and influencing factors. The geographical detector model (Geodetector) is a statistical method that can detect spatial differentiation and reveal the driving forces behind it. Further, it can reveal the linear and nonlinear relationships between environmental pollution and influencing factors [24]. Therefore, this study uses Geodetector to analyze the factors affecting the spatial distribution of NO₂ in the YRD.

This study is based on the NO₂ column concentration data gathered by the TROPOMI on the Sentinel-5P. The spatial–temporal distribution characteristics of the NO₂ column concentration in the YRD from 2019 to 2020 were analyzed using the Google Earth Engine (GEE) platform, and Geodetector was used to analyze the driving factors of the NO₂ column concentration, providing a scientific basis for atmospheric environmental governance.

2. Materials and Methods

2.1. Study Area

The YRD includes 41 cities in Shanghai, Jiangsu Province, Zhejiang Province, and Anhui Province (Figure 1). It is located in the lower reaches of the Yangtze River, adjacent to the Yellow Sea and the East China Sea. It is an alluvial plain that is formed before the Yangtze River enters the sea. The entire region experiences a typical subtropical monsoon climate, which is hot and rainy in summer and cold and dry in winter. The total area is 358,000 km²; by the end of 2020, the permanent resident population had reached 235 million people. It is a region with one of the best urbanization foundations in China. However, with increases in its population, the intensive industrialization of its cities, and increases in economic activity, its atmospheric environment has been severely affected, harming human health and stifling social and economic progress.

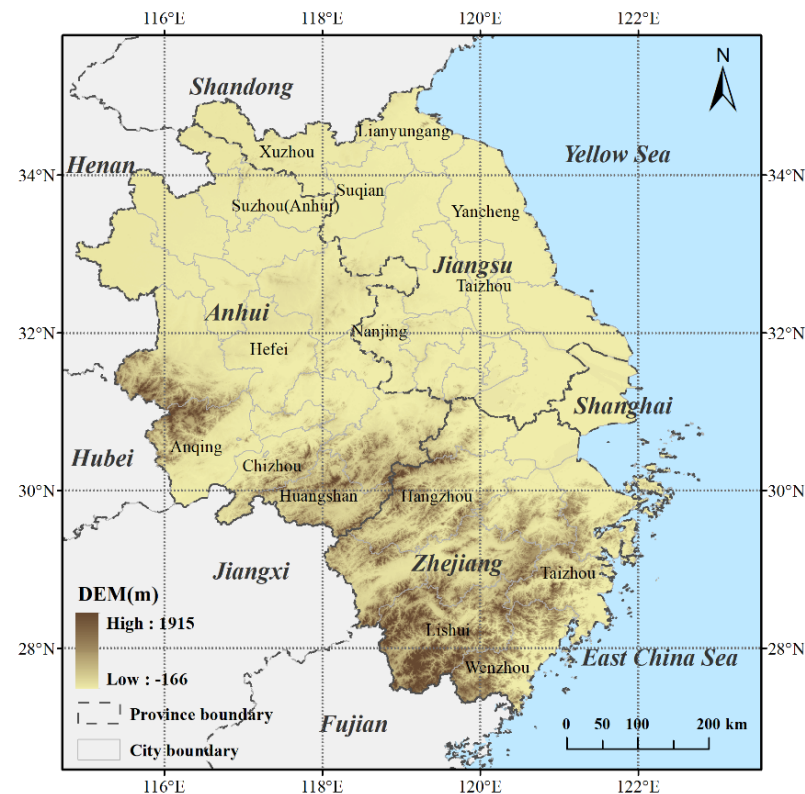


Figure 1. Overview of the study area (note: we used elevation data as a baseline to create the map of the study area; DEM can realize the digital simulation of ground terrain with limited terrain elevation data and express the topographic conditions of the YRD. In the study area map, we have labeled the provincial and municipal boundaries).

2.2. Data Sources

The following data were used in this study (Table 1). The NO₂ column concentration data observed by the TROPOMI sensor mounted on the Sentinel-5P satellite were used. The NO₂ column concentrations (unit: mol/m²) are the summations of the atmospheric NO₂ from the ground to the top of the troposphere (approximately 10 km in the mid-latitudes) [25]. They were obtained by a series of inversion algorithms applied to the data observed by satellite sensors [26,27]. Based on the GEE platform, the images of the YRD from

January 2019 to December 2020 were obtained, and the mean value per month was synthesized. A total of 24 NO₂ column concentration images were generated for subsequent analysis.

Table 1. Primary data used in this study.

Variable	Indicator	Abbreviation	Source
NO ₂	NO ₂ column concentrations	–	Google Earth Engine
	Ground-monitored NO ₂ concentrations	–	China National Environmental Monitoring Centre
Physical geographical factors	Normalized difference vegetation index	NDVI	Google Earth Engine
	Altitude	DEM	
	Temperature	TEMP	China Statistical Yearbook
	Precipitation	PRCP	
Socio-economic factors	Per capita gross domestic product	PGDP	China Statistical Yearbook
	Secondary industry	SI	
	Civil car ownership	CAR	

The in situ observed NO₂ concentrations at ground stations for air quality monitoring in the YRD were acquired from the China National Environmental Monitoring Centre. In this paper, the monthly average NO₂ concentrations of 41 cities in the YRD region from 2019 to 2020 were obtained as a correlation test to verify the applicability of the NO₂ column concentration data.

We selected seven influencing factors. (1) The normalized differential vegetation index: the spatial resolution of the NDVI dataset is 1 km, with a period of 16 days, derived from the 16-day synthetic product ‘MOD13A1v006’ [28]. (2) Altitude: the spatial resolution of the DEM dataset is about 30 m, derived from ‘SRTM Digital Elevation Data’ [29]. The NDVI and DEM datasets were resampled to 3.5 km × 3.5 km, with the NO₂ column concentration as the basic reference. Other data included the temperature (TEMP), precipitation (PREC), regional gross domestic product (PGDP), total secondary industry (SI), and civilian vehicle ownership (CAR) of the entire area, obtained from the National Statistical Database and the provincial and prefectural Statistical Yearbooks of 2019 and 2020.

2.3. Methods

2.3.1. Spatial Autocorrelation Analysis

In this study, we analyzed the spatial agglomeration characteristics of NO₂ column concentrations in the YRD region in 2019 and 2020. The most commonly used methods to accomplish this are global spatial autocorrelation and local spatial autocorrelation; these can calculate the characteristics of spatial aggregation (high-value aggregation/low-value aggregation) according to the correlations of spatial objects [29–31]. Spatial autocorrelation is widely used in the field of geography, which can determine whether spatial objects are related. A high correlation proves that there is a spatial agglomeration phenomenon [32].

The global spatial autocorrelation is represented by Moran’s *I* index. It can be used to describe the overall distribution of NO₂ column concentrations in the YRD and to ascertain whether there is an agglomeration phenomenon in space. The calculation of Formula (1) is

$$\text{Moran's } I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{y})(x_j - \bar{y})}{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (1)$$

where $n = 41$ is the number of cities in the YRD. x_i and x_j are the NO₂ column concentration of cities i and j , respectively; $\bar{y}$ is the average concentration of NO₂ in all cities; and w_{ij} is a spatial adjacency weight matrix. If Moran’s *I* > 0, a positive spatial correlation exists. The larger the value, the more obvious the correlation is; if Moran’s *I* < 0, a negative spatial correlation exists, and the smaller the value, the greater the spatial difference. Finally, if Moran’s *I* = 0, there is no correlation [33].

Local spatial autocorrelation is represented by the local indicators of spatial association (LISA) [34]. This method can analyze attribute information in local areas to reflect local autocorrelation and spatial aggregation. In this paper, LISA is used to represent the accumulation characteristics of NO₂ column concentration in the YRD. The calculation of Formula (2) is

$$LISA = \frac{(x_i - \bar{x}) \times \sum_{j=1}^n W_{ij}(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (2)$$

where the variables have the same meaning as in Formula (1).

2.3.2. Hotspot Analysis

To make the spatial characteristics of the statistics more convincing, in this study, hotspot analysis was used to statistically calculate the areas of spatial hotspots (high-value aggregation) and spatial coldspots (low-value aggregation) [35,36]. Spatial hotspot analysis can detect abnormal areas with certain attribute values that significantly differ from those of other areas in the research area [36]. It is advantageous to quantitatively analyze the high–low agglomeration areas of NO₂ column concentrations from the perspective of spatial statistics to determine the evolution of the pollution agglomeration area and assist in the adjustment of the regional joint prevention and control policy. Firstly, the spatial hotspot detection model constructs a symmetric matrix according to the spatial distances corresponding to all samples in the research area, and then determines the aggregation area according to the limit distance method. The calculation of Formula (3) is

$$D = 0.5\sqrt{A/n} \pm \gamma \left(\frac{0.26136}{\sqrt{n^2/A}} \right) \quad (3)$$

where D is the limit distance, km; A is the area of the YRD, km²; n is the number of cities in the YRD; and γ is the quantile when the confidence level is given.

2.3.3. Geodetector

When analyzing the factors affecting the NO₂ column concentration, the OLS, GWR [17], SLM, and SEM [23] methods can be used. These methods only express the linear relationship between independent variables and dependent variables, but the geodetector can express the linear and nonlinear relationships between the NO₂ column concentration and the influencing factors [37]; therefore, it is used for the influencing factor analysis of the NO₂ column concentration.

Geodetector is a statistical method used to detect the spatial heterogeneity of parameters and the driving factors behind this heterogeneity, originally developed by Wang et al. [24]. This includes four main functions: factor detector, interaction detector, risk detector, and ecological detector. This paper uses the factor detector and interaction detector to study the degree of explanation of each factor in the process of the spatial differentiation of NO₂ column concentration.

The factor detector can detect the spatial heterogeneity of NO₂ column concentration and the extent to which a certain factor X explains this heterogeneity, which is measured by the q value, whose expression is given in Formula (4):

$$q = 1 - \frac{\sum_{h=1}^L N_h \sigma_h^2}{N \sigma^2} = 1 - \frac{SSW}{SST} \quad (4)$$

where $h = 1, \dots, L$ is the stratification of the NO₂ column concentration or factor X, i.e., the classification or partition; N_h and N are the number of layers h and area units, respectively; and σ_h^2 and σ^2 are the variances of the values of layer h and the entire region's NO₂ column concentration, respectively. SSW and SST are the sums of the intra-layer variance and regional total variance, respectively. The value range of q is (0, 1); the larger the value, the more obvious the spatial differentiation of NO₂ column concentrations. If the stratification is generated by

the independent variable X , the larger the q value, the stronger the explanatory power of the independent variable X for the NO_2 column concentration, and vice versa.

Further, interaction detection can identify the interactions between different risk factors XS , evaluate whether factors $X1$ and $X2$ will increase or decrease the explanatory power of the NO_2 column concentration when used in combination, or whether the influence of these factors on the NO_2 column concentration is independent. The method of evaluation involves first calculating the explanatory power q values of $X1$ and $X2$ for the NO_2 column concentration: $q(X1)$ and $q(X2)$. The q values are then calculated when they interact. The main data processing and analysis methods used in this study are summarized in Figure 2.

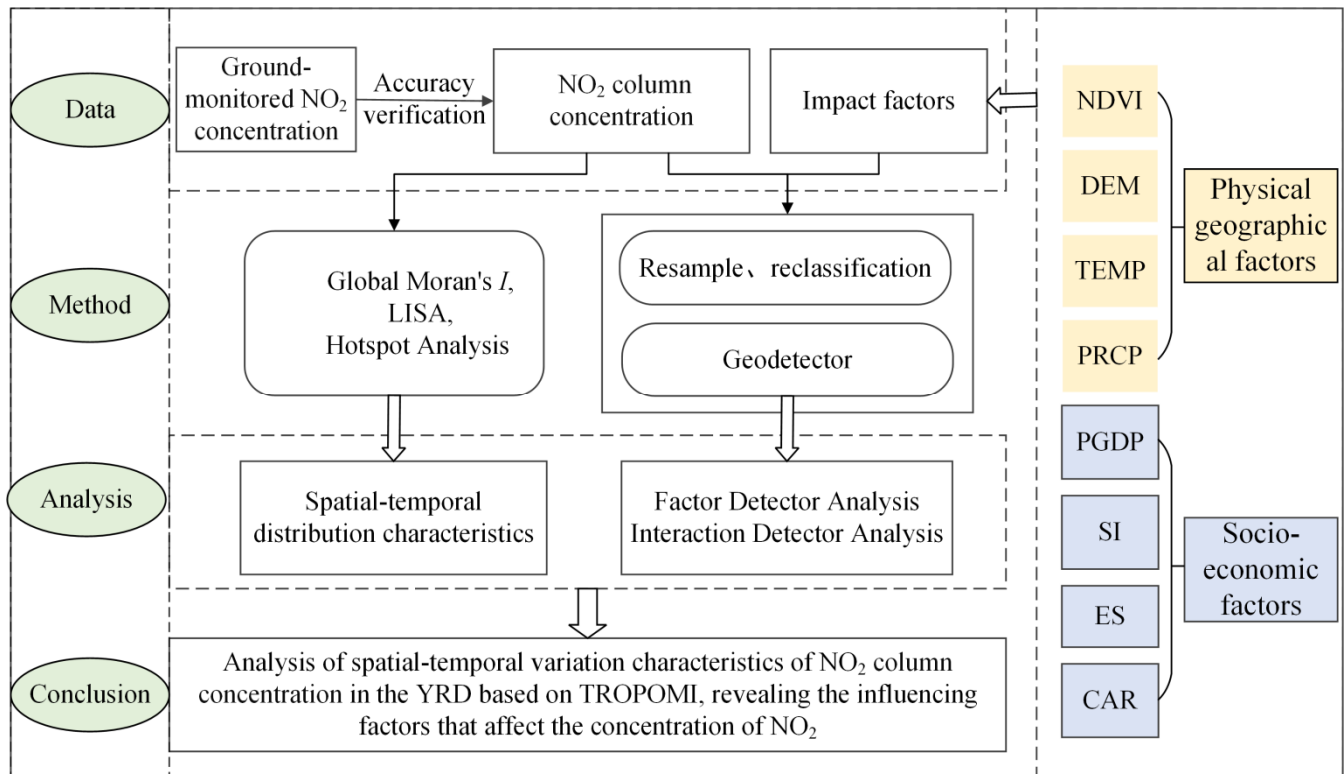


Figure 2. Flowchart of the research methodology.

3. Results

3.1. Comparison and Verification between the NO_2 Column Concentration and the Ground-Monitored NO_2 Concentrations

At present, some studies have confirmed that the NO_2 column concentration has a strong correlation with the ground-monitored NO_2 concentrations [16,17,38–40]. Further, Judd et al. [14] showed that the NO_2 column concentration inversion by TROPOMI is underestimated to a certain extent, and the inversion results in a large-scale range are 85% consistent with ground-monitoring data. In this study, to verify the accuracy of the data, linear fitting was performed on the monthly average NO_2 ground-monitored concentration data and NO_2 column concentration data in the YRD region from 2019 to 2020 (Figure 3). The results show that the correlation between the two reached 70%, and the concentration distribution trend remains basically the same; therefore, the correlation between the two is strong, and the NO_2 product retrieved by TROPOMI can stably reflect the actual NO_2 pollution on the surface.

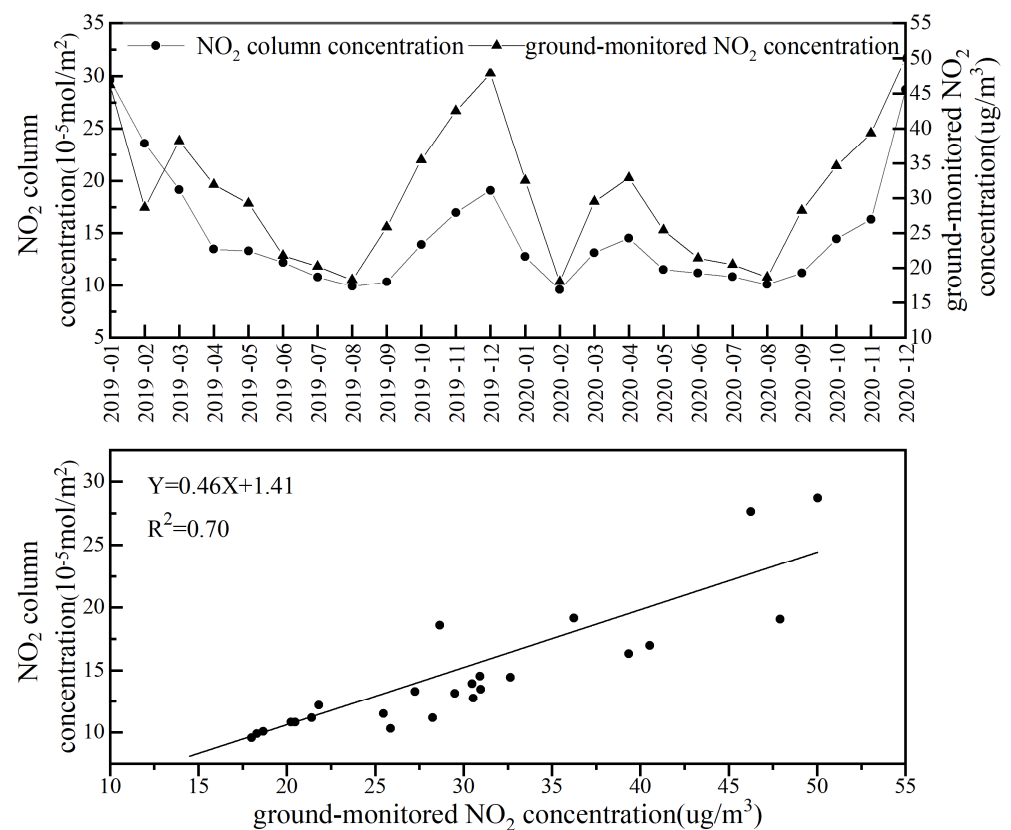


Figure 3. Correlation analysis of the ground-monitored NO_2 concentrations and NO_2 column concentration in the YRD.

3.2. Spatial–Temporal Distribution Characteristics of the NO_2 Column Concentration

3.2.1. Temporal Distribution Characteristics of the NO_2 Column Concentration in the YRD

This study calculates the monthly mean NO_2 column concentration in the YRD region from January 2019 to December 2020 (Figure 4). In 2019, the annual variation trend of the NO_2 column concentration exhibited a ‘U’-shaped curve, with ‘high in winter and low in summer, with a transition between spring and autumn’ characteristics. The maximum concentrations appeared in January, with an average value of $30 \times 10^{-5} \text{ mol/m}^2$. Then, the concentration gradually decreased, reaching a minimum in August, with a concentration value of $10 \times 10^{-5} \text{ mol/m}^2$; finally, the NO_2 column concentration gradually rose from September to December. In comparison to the same period in 2019, the temporal distribution characteristics of the NO_2 column concentration in each month in 2020 were consistent on the whole, but there were obvious changes in different months. Specifically, the NO_2 column concentration in January, February, and March of 2020 decreased by 56.91%, 59.42%, and 31.4%, respectively, in comparison to the levels in 2019; however, the concentration in December increased significantly.

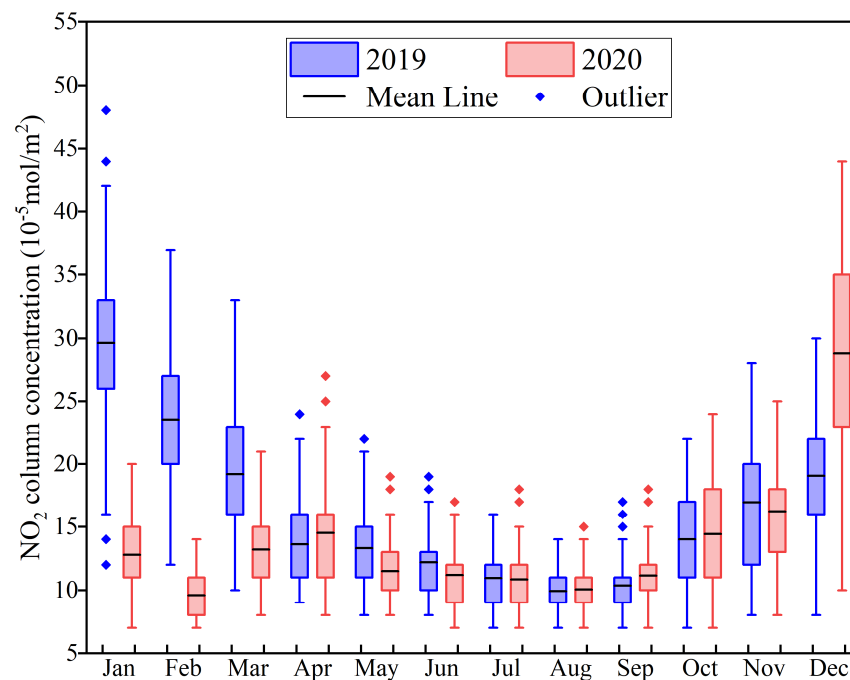


Figure 4. Distribution characteristics of the monthly average of NO_2 column concentration in the YRD from 2019 to 2020 (note: this is a violin plot showing the monthly mean change trend. The rectangular box corresponds to the 25–75% range of the NO_2 column concentration, red and blue diamonds represent outliers).

3.2.2. Spatial Distribution Characteristics of NO_2 Column Concentration in the YRD

Our research obtained the spatial distribution of the 24-month average NO_2 column concentration remote sensing images of the YRD region from January 2019 to December 2020 (Figure 5). These data were divided into 10 levels (the data in January and February have slight portions missing in Hefei, Shaoxing, and Shanghai, but this does not affect the overall analysis). The cities with the highest concentrations of NO_2 in the YRD are located in the ‘Hefei–Nanjing–Wuxi–Suzhou–Shanghai’ region, and are distributed in a striped formation; this is followed by the concentration in the northern region. The areas with the lowest concentrations are located in the southwest mountainous areas, represented by the ‘Huangshan–Chuzhou–Lishui–Wenzhou–Taizhou’ region in the south and the ‘Yancheng’ region in the northeast. The spatial distribution map of the NO_2 column concentration shows that the concentration in winter (December–February) was the highest, and the concentration in summer (June–August) was the lowest, with slight differences between spring and autumn. Comparing the concentration distribution for 12 months in 2019 and 2020, it can be seen that there are obvious changes in January, February, March, and December. In 2020, the concentration in January and February decreased significantly in the entire region, and the concentration began to increase in March; however, it was still lower than that in 2019, and the concentration in Jiangsu Province and northern Anhui Province increased significantly in December.

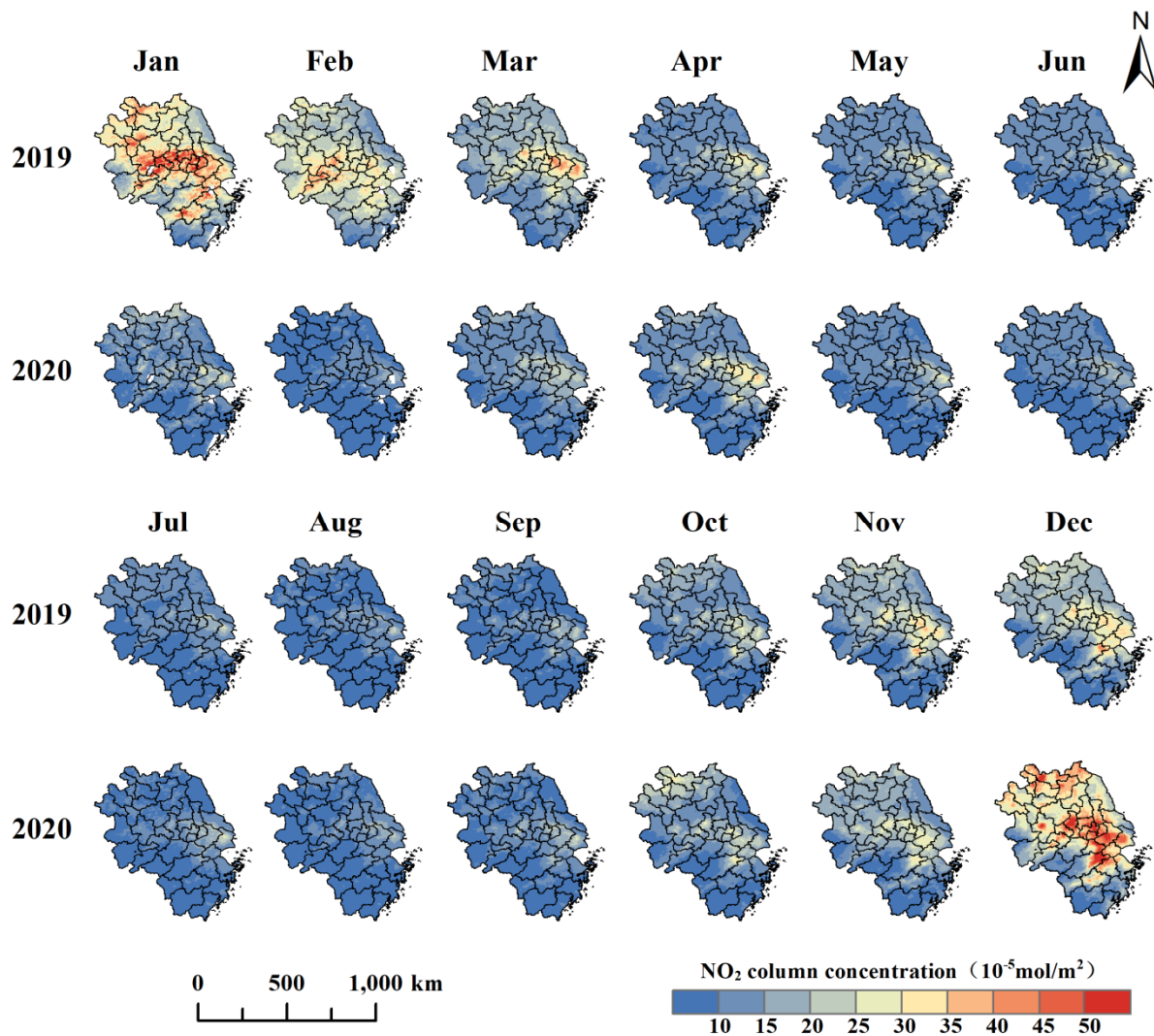


Figure 5. Monthly spatial distribution of the NO₂ column concentration in the YRD from 2019 to 2020.

3.3. Spatial Agglomeration Characteristics of the NO₂ Column Concentration in the YRD

In our study, a 10×10 km grid was constructed in the YRD region to extract the NO₂ column concentration value. Next, the global spatial autocorrelation of the 2019 and 2020 annual average NO₂ column concentrations was analyzed. The Moran's I value of the NO₂ column concentrations in 2019 and 2020 were 0.776 and 0.797, respectively; the z values were all greater than 2.58; and the p values were all less than 0.01. This shows that there was a very significant positive correlation in the spatial distribution of the NO₂ column concentration in the YRD region, and there was an obvious spatial agglomeration characteristic. Therefore, to determine the regions of high and low NO₂ column concentrations, it was also necessary to analyze the characteristics of local spatial autocorrelation.

Local spatial autocorrelation can intuitively express the characteristics of the NO₂ column concentration local spatial distribution (Figure 6a,b). Further, 99% of the regions belong to three types: high–high agglomeration, low–low agglomeration, and no significant agglomeration, all of which pass the 95% significance test. High–high agglomeration regions are mainly located in the central region of the YRD, including central cities such as Hefei, Shanghai, Nanjing, and Hangzhou, and northern cities such as Xuzhou and Lianyungang. Low–low agglomeration regions are concentrated in Yancheng and cities with higher altitudes in the south, including Anqing, Chizhou, Huangshan, Quzhou, Lishui, Wenzhou, and Taizhou.

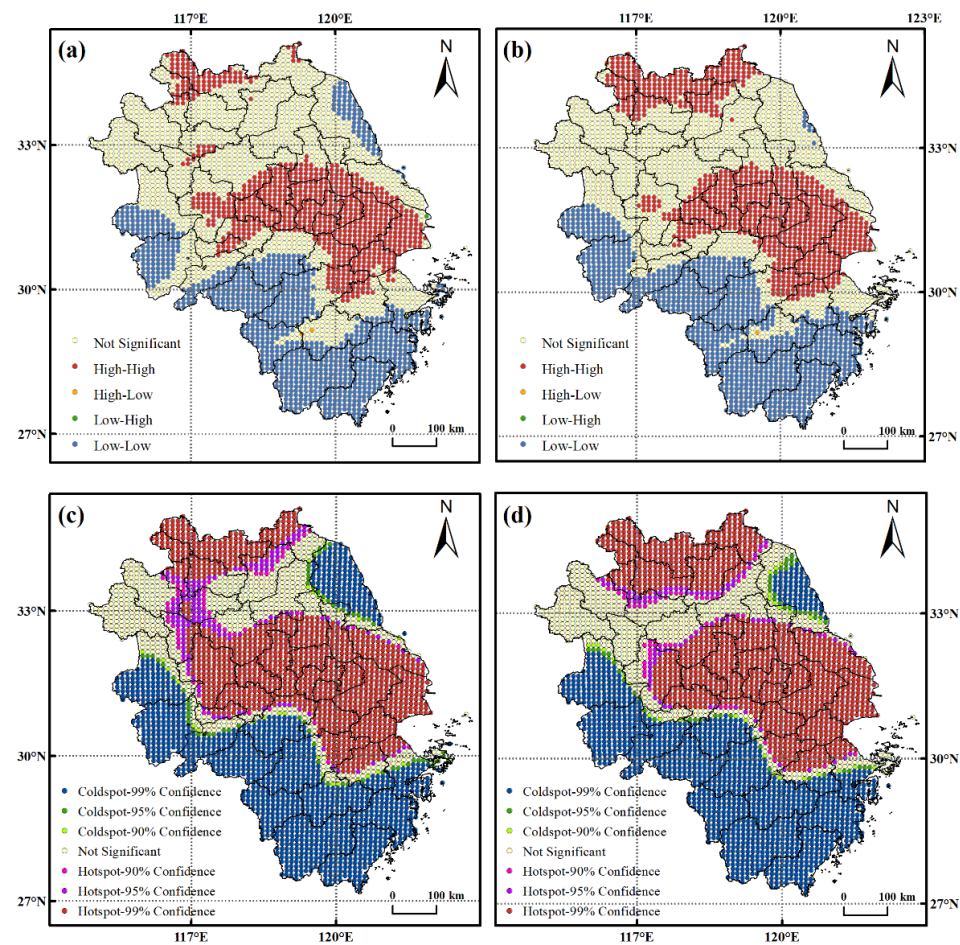


Figure 6. Characteristics of the NO_2 column concentration spatial agglomeration. (a) LISA agglomeration map of NO_2 column concentration in 2019. (b) LISA agglomeration map of NO_2 column concentration in 2020. (c) Hotspot analysis in 2019. (d) Hotspot analysis in 2020.

In this study, the Euclidean distance weight method was used to carry out a spatial agglomeration analysis of the NO_2 column concentrations in the YRD region. In 2019 and 2020, both the coldspot and hotspot agglomeration areas exhibited a flake-like distribution (Figure 6c,d). The aggregated areas of hotspots decreased in general, among which the radiation area of hotspots from Hefei to Suzhou gradually decreased; however, the aggregated areas of hotspots from Huai'an and Suqian increased. The aggregated area of coldspots generally increased gradually, radiating from the southwest to the center of the YRD; however, the aggregated areas of coldspots in Yancheng gradually decreased.

The results of both analysis methods describe the state of aggregation of high and low values. However, due to the different statistical methods used, the results differ. In this study, two methods were used to judge the area of high-value aggregation and low-value aggregation, and both results show that high-value aggregation areas are mainly distributed in the central region of the YRD, and low-value aggregation areas are mainly in Yancheng City and high-terrain areas in the south.

3.4. Influencing Factor Analysis of NO_2 Column Concentration Distribution in the YRD

According to the extant studies, the emission sources of NO_2 are divided into two types: natural emissions and anthropogenic emissions [22,41]. Therefore, we selected the following seven influencing factors of the NO_2 column concentration distribution in the YRD in 2019 and 2020 for analysis using Geodetector: the normalized difference vegetation index (NDVI), altitude (DEM), temperature (TEMP), precipitation (PRCP), per capita gross domestic product (PGDP), the secondary industry (SI), and civil car ownership (CAR). First,

the value of each factor was discretized, and then each factor was graded using the natural discontinuity method, and the processed data were sorted and imported into Geodetector to generate the results.

3.4.1. Factor Detector Analysis

According to the factor interpretation power analyzed using Geodetector, the influences of different factors on the spatial-temporal distribution of the NO₂ column concentrations differed (Table 2). In 2019, the driving factors affecting the spatial-temporal distribution of NO₂ were ranked as follows: $q(\text{NDVI}) > q(\text{DEM}) > q(\text{TEMP}) > q(\text{PGDP}) > q(\text{CAR}) > q(\text{SI}) > q(\text{PRCP})$. In 2020, the influence ranking of driving factors was as follows: $q(\text{NDVI}) > q(\text{DEM}) > q(\text{PGDP}) > q(\text{TEMP}) > q(\text{CAR}) > q(\text{SI}) > q(\text{PRCP})$. The influences of NDVI and DEM were the largest, both exceeding 0.6, and the influence of other factors exceeded 0.35. By comparing the influences of various factors in 2019 and 2020, it can be seen that the q value of the PGDP, the CAR, and the ES increased, indicating that economic development and traffic had a greater impact on the concentrations of the NO₂ column concentration. Further, the q values of NDVI, DEM, and PRCP remained stable, and the q values of TEMP and SI decreased slightly.

Table 2. The q values and p values of potential impact factors in 2019 and 2020.

Influence Factor	2019	2020
	q Value	q Value
NDVI	0.746 *	0.741 *
DEM	0.692 *	0.695 *
TEMP	0.482 *	0.425 *
PRCP	0.359 *	0.351 *
PGDP	0.479 *	0.553 *
SI	0.370 *	0.351 *
CAR	0.384 *	0.421 *

Note: The q value represents the explanatory power of the factor, and the value range is (0, 1). The larger the value, the stronger the explanatory power of NO₂ column concentration. An asterisk (*) indicates that the influencing factors are significant at the 1% level.

In addition, we used the Pearson correlation coefficient to indicate the direction of the influence of detection. The results are shown in Figure 7: the NDVI, DEM, temperature, and precipitation are negatively correlated with the NO₂ column concentration, while the per capita GDP, secondary industry, and car ownership are positively correlated with the NO₂ column concentration.

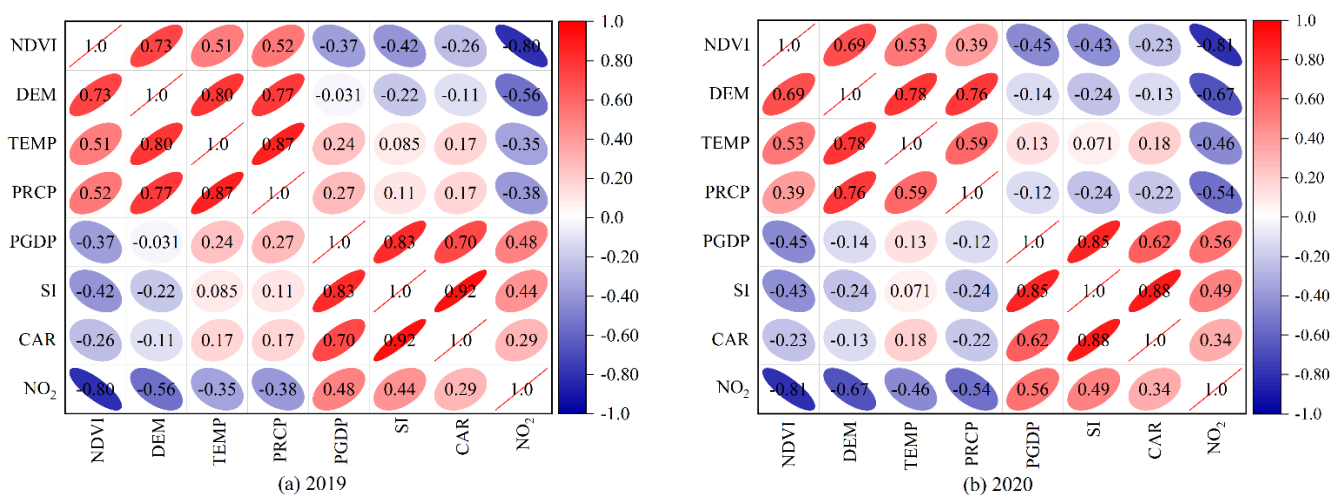


Figure 7. Pearson correlations of the NO₂ column concentration and its potential impact factors of the YRD in 2019 and 2020.

3.4.2. Interaction Detector Analysis

The interactions among factors can be used to ascertain whether the combined effect of different influencing factors on the dependent variable is enhanced, weakened, or independent. This study found that each factor had an interaction effect on the NO₂ column concentration, and under the superposition of any two factors, there were two kinds of relations: double-factor enhancement and nonlinear enhancement (Figure 7). Among these relationships, double-factor enhancement accounted for the majority, i.e., factor superposition has a strong driving effect on the concentration changes of NO₂. Only a small number of factors, such as precipitation and other factors, exhibited nonlinear enhancement.

By comparing the results of the factor interaction detection in 2019 and 2020, it can be seen that the interaction q value of each factor was mostly greater than 0.8. Further, the interaction force between socio-economic factors and other factors in 2020 was stronger than that in 2019, indicating that the impact of economic development on NO₂ concentrations increased. Overall, the influence of various factors on changes in the NO₂ concentrations was not independent but had significant interactions. The influence of the interaction of multiple factors on changes in the NO₂ concentrations was not a simple superposition process, but a double-factor enhancement or nonlinear enhancement.

4. Discussion

4.1. Spatial–Temporal Distribution Characteristics of NO₂ Column Concentration

The NO₂ column concentration in the YRD region was high in winter and low in summer in 2019; this was determined by many factors. (Figures 4 and 5). The vegetation coverage, temperature, and precipitation in winter were significantly lower than those in summer; therefore, the NO₂ pollutants that gathered over the city were not easily transformed, decomposed, and dissipated. For this reason, the concentration of NO₂ in winter was higher than that in summer. This is basically consistent with the monthly variation trend of the NO₂ column concentrations found in Zhou's study [41]. In comparison to 2020, the overall changes in the NO₂ column concentration basically remained the same; however, the changes in the individual months were significant. In January and February, owing to COVID-19 restrictions, industrial production basically ceased, and traffic source emissions were also restricted, resulting in a significant reduction in the NO₂ column concentration [42,43]. In March, work and production resumed, citizens could wear masks for outdoor activities, and short-distance travel was not restricted. Therefore, the concentration of NO₂ began to increase slowly. In April, the concentration of NO₂ basically returned to the pre-pandemic level, which indicates that the impacts on the NO₂ column concentration were relatively short-term and could gradually return to normal levels after the implementation of the release policy. Notably, the NO₂ column concentration in December 2020 was higher than that in the same period in 2019, which may be due to the influence of unfavorable meteorological conditions (such as low wind and temperature inversion). According to the report released by the Department of Ecology and Environment of Jiangsu Province on 11 December 2020 [44], the air quality of cities along the Yangtze River in Jiangsu Province reached the level of mild to moderate pollution. The atmospheric diffusion conditions in Jiangsu Province further deteriorated, resulting in heavy pollution. Marco et al. [45] showed that pollutant concentrations in the atmosphere are controlled not only by emissions but also by meteorological processes. As a consequence, adverse atmospheric conditions may hinder the effects of policies intended to improve air quality through the reduction of emissions. In particular, low ventilation conditions and temperature inversions could significantly inhibit pollutant transport and mixing, leading to high concentrations close to the ground.

This study analyzed the aggregation characteristics of the NO₂ column concentrations in the YRD region (Figure 6), finding that areas with high–high-value aggregations are located in the central cities of the YRD, and areas with low–low-value aggregations are located in northern Yancheng and southern areas. We used the spatial distribution map of the 2019 NO₂ column concentrations and various influencing factors to conduct an auxiliary

analysis (Figure 8). The central part of the YRD exhibited high per capita GDP, a rapid development of secondary industries, and increased car ownership, resulting in an increase in anthropogenic emissions. The vegetation coverage was poor, the sources of NO₂ were higher than the consumption, and a high NO₂ column concentration value resulted. As a coastal city in the YRD, Yancheng City has slower economic development and fewer anthropogenic emissions than other coastal cities, such as Nantong and Shanghai; therefore, the low NO₂ column concentration value has become concentrated. The vegetation coverage in the southern region is relatively high, and a good ecological environment can alleviate air pollution to a certain extent, lowering the NO₂ column concentration.

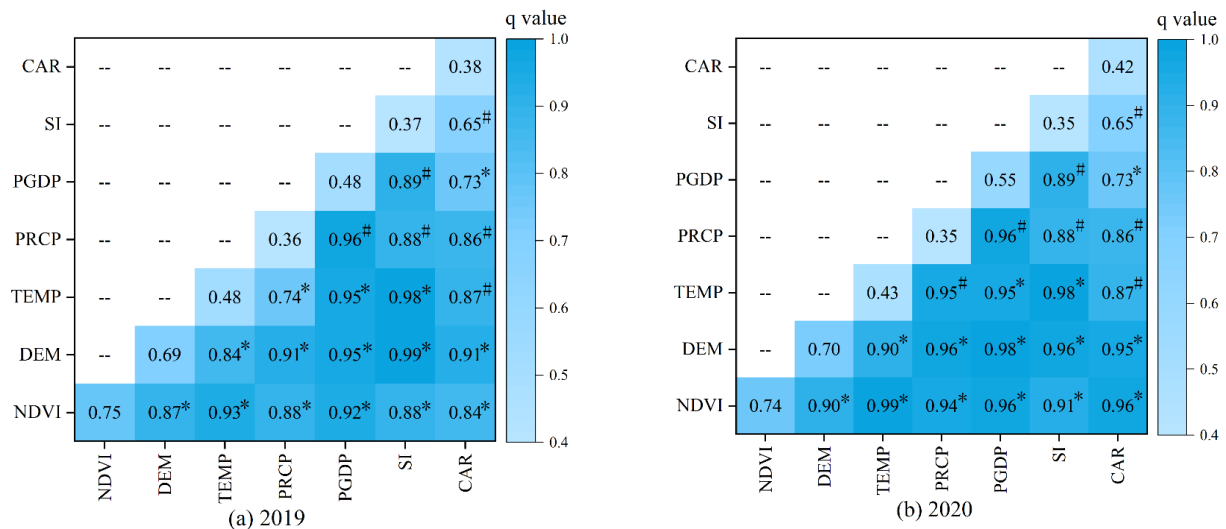


Figure 8. Interactive enhancement of potential impact factors on NO₂ column concentrations of the YRD in 2019 and 2020 (if $q(X1 \cap X2) > q(X1) + q(X2)$ represents nonlinear enhancement, it is denoted by the octothorpe (#); if $q(X1 \cap X2) > \text{Max}(q(X1), q(X2))$ represents double-factor enhancement, it is denoted using an asterisk (*)).

4.2. Influencing Factors of NO₂ Column Concentration

4.2.1. Physical Geographical Factors

Meteorological factors play an important role in influencing the concentration of NO₂ in the atmosphere. Precipitation has the effect of removing and scouring atmospheric pollutants, while temperature affects solar radiation, which directly affects the photolysis effect of NO₂ in the atmosphere [46–49]. From a temporal perspective (Figure 5), temperature and precipitation lead to monthly fluctuations in NO₂ concentration. Winter is mild and less rainy, resulting in high NO₂ concentrations, while summer is hot and rainy, with lower NO₂ concentrations. From a spatial perspective (Figure 9a,b,c), the southern regions, with higher temperatures and precipitation, have lower NO₂ concentrations, while the opposite is true in the northern regions.

From the detection results, it is clear that the NO₂ column concentration in the YRD is most affected by vegetation coverage. Xiao et al. [22] showed that vegetation cover has a certain absorption and adsorption effect on NO₂ in local areas. Plants absorb NO₂ and combine with water attached to the surface of sponge tissue to generate nitrous acid or nitric acid, which accumulates in plants; plants adsorb nitrate particles in the air and improve the efficiency of atmospheric NO₂ conversion to nitrate particles. Therefore, the higher the vegetation coverage, the lower the NO₂ column concentration. This is consistent with our findings that the NO₂ concentration was negatively correlated with vegetation cover in the YRD, and that the NO₂ was lower in areas with higher vegetation coverage in the south; the opposite was true in the central region.

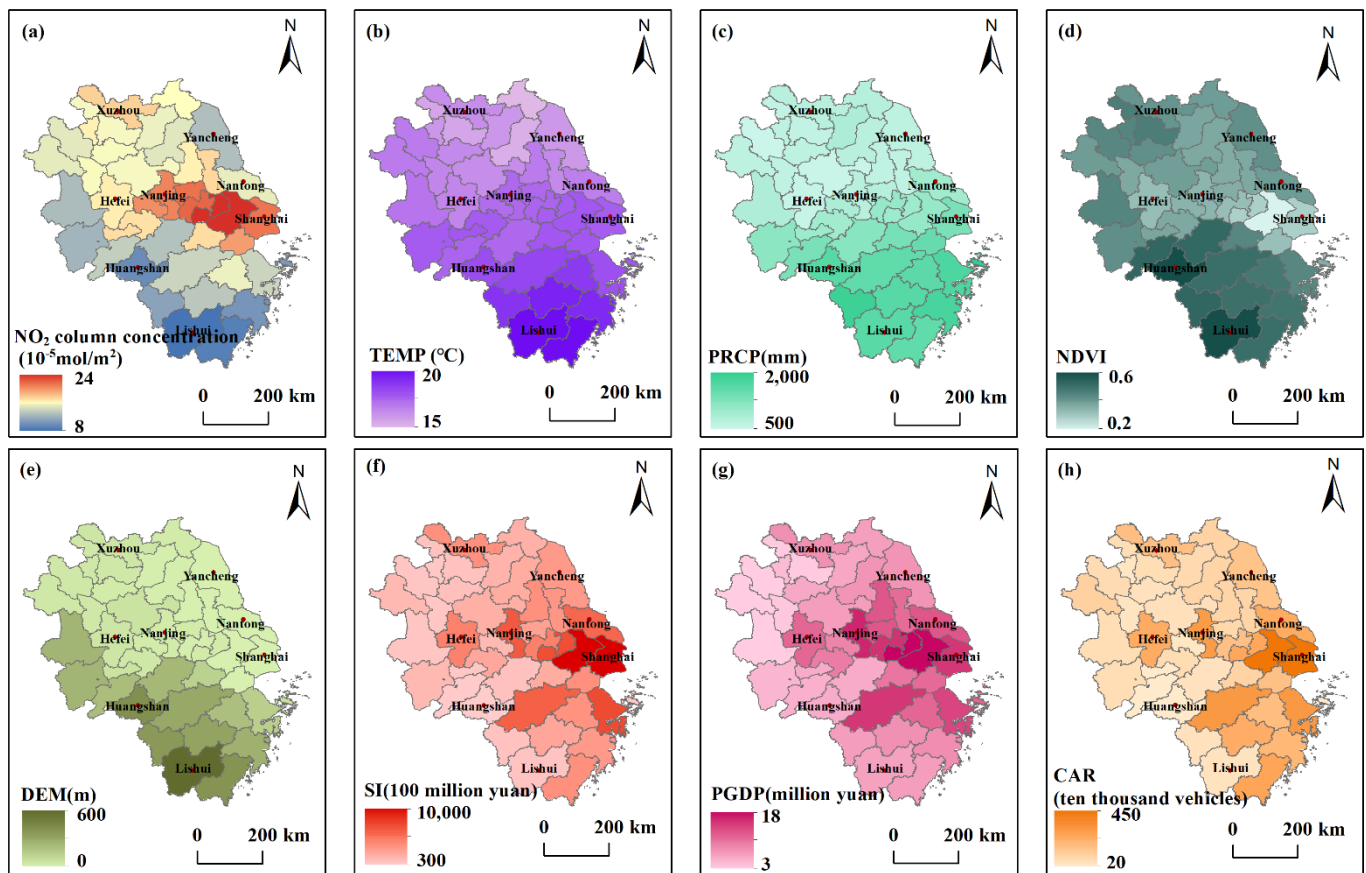


Figure 9. Spatial distribution map of NO₂ column concentration and influencing factors in 2019. (notes: (a) NO₂ concentration distribution; (b) Temp: Temperature distribution; (c) Prec: Precipitation distribution; (d) NDVI: Normalized vegetation index; (e) DEM: Digital Elevation Model; (f) SI: Secondary industry; (g) PGDP: regional gross domestic product; (h) CAR: civilian vehicle ownership.)

The explanatory power of the altitude for NO₂ column concentration was 0.69 (Figure 7). The influence of altitude on the change of NO₂ is divided into two aspects: on the one hand, high altitudes weaken the diffusion of the atmosphere from the central part of the YRD to the south [17], such that NO₂ accumulates in the central region; on the other hand, Brooke et al. [50] showed that altitude is positively correlated with vegetation growth, which indirectly affects NO₂. It can be seen that the spatial distribution of altitude and vegetation growth was basically consistent (Figure 9d,e), which also verifies our assertions.

4.2.2. Socio-Economic Factors

In terms of secondary industries, the combustion of fossil fuels and the application of other energy sources are significant sources of air pollution. Therefore, upgrading and adjusting the industrial structure is an effective measure to control air pollution [51]. According to the statistical yearbooks of the four provinces in the YRD region, the energy consumption of industrial enterprises is dominated by raw coal and coke. Areas with high secondary industry output values also have high NO₂ column concentrations (Figure 9a,f). In addition, we used the per capita GDP to represent the economic level of the region. Economic development will inevitably consume energy, lead to increased waste materials, and cause air pollution [52]. Areas with better economic development in the YRD also have higher NO₂ column concentrations (Figure 9a,g).

Economic growth will indirectly increase civil car ownership, and this increase will directly increase the NO₂ concentrations. The main pollutants produced by vehicle exhaust emissions include nitrogen oxides, hydrocarbons, and fine particles, which have become the main source of air pollution in China [46,53–55]. Cities such as Hefei, Nanjing, Suzhou,

Shanghai, and Hangzhou, with a large number of cars in the YRD, suffer from severe NO₂ pollution (Figure 9a,h), while cities such as Lishui, Quzhou, and Huangshan have less car ownership, with a better atmospheric environment.

4.3. Strengths and Limitations of this Study

This study used the GEE cloud platform to process and quickly obtain the NO₂ column concentration image data of the YRD from 2019 to 2020. These data can objectively reflect the spatial evolution trends of NO₂ column concentrations in the YRD and provide a reference for research on air pollution control.

However, there are certain limitations of this study. Firstly, because the Sentinel-5P data were from July 2018, the long-term series findings of this study require further analysis. In addition, even though we selected the influencing factors as comprehensively as possible, this selection could always be improved. Therefore, in future research, we will consider using multi-source data for analysis to study the characteristics of long-term NO₂ changes and fully consider the driving factors affecting NO₂, which could facilitate a deeper understanding of NO₂ column concentrations.

5. Conclusions

In this paper, the spatial–temporal distribution characteristics and influencing factors of NO₂ column concentrations in the YRD were analyzed. The following conclusions can be drawn from the results:

- (1) Based on the ground-monitored NO₂ concentration data, the correlation tests showed that the NO₂ column concentration observed by TROPOMI could reflect the real NO₂ pollution scenario on the surface.
- (2) The annual variation trend of NO₂ column concentrations in the YRD region from 2019 to 2020 exhibited a ‘U’-shaped curve, and there was a seasonal characteristic of ‘high in winter and low in summer’. In terms of the spatial distribution, the NO₂ column concentration in the YRD was the highest in the central region and the lowest in high-altitude areas in the south and the coastal area in the northeast.
- (3) The factor detection results show that the influences of vegetation and altitude were the largest. Further, by comparing the influences of various factors in 2019 and 2020, it was found that economic development and traffic had a greater impact on the concentrations of NO₂.

Author Contributions: Conceptualization, Z.Z.; methodology, X.G.; software, X.G.; validation, Z.Z.; formal analysis, X.G. and Z.Z.; data curation, Z.Z.; writing—original draft preparation, X.G.; writing—review and editing, X.G., Z.Z. and Z.C.; visualization, X.G., Y.X. and J.Z.; supervision, Z.Z.; project administration, L.W. and Z.G.; funding acquisition, Z.Z. and L.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China (Grant Nos. 42071085 and 41701087), the Open Project of the State Key Laboratory of Cryospheric Science (Grant No. SKLCS 2020-10), and the Graduate Scientific Research Project of Anhui Universities in 2021 (Grant No. YJS20210412).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Sentinel-5P NO₂ data, “MOD13A1v006” and “SRTM Digital Elevation Data” were obtained from the Google Earth Engine platform: <https://developers.google.com/earth-engine/datasets/catalog> (accessed on 1 March 2022). Ground-monitored NO₂ concentrations were acquired from the China National Environmental Monitoring Centre. Other influencing factor data were acquired from the National Statistical Database and the provincial and prefectural Statistical Yearbooks of 2019 and 2020.

Acknowledgments: We want to express our sincere gratitude to the anonymous reviewers and editors for their efforts in the improvement of the paper.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Cui, Y.; Wang, L.; Jiang, L.; Liu, M.; Wang, J.; Shi, K.; Duan, X. Dynamic spatial analysis of NO₂ pollution over China: Satellite observations and spatial convergence models. *Atmos. Pollut. Res.* **2021**, *12*, 89–99. [CrossRef]
2. Zhang, X.; Zhang, P.; Zhang, Y.; Li, X.; Qiu, H. The trend, seasonal cycle, and sources of tropospheric NO₂ over China during 1997–2006 based on satellite measurement. *Sci. China Ser. D Earth Sci.* **2007**, *50*, 1877–1884. [CrossRef]
3. Chen, R.; Huang, W.; Wong, C.-M.; Wang, Z.; Quoc Thach, T.; Chen, B.; Kan, H. Short-term exposure to sulfur dioxide and daily mortality in 17 Chinese cities: The China air pollution and health effects study (CAPES). *Environ. Res.* **2012**, *118*, 101–106. [CrossRef] [PubMed]
4. Cui, Y.; Zhang, W.; Bao, H.; Wang, C.; Cai, W.; Yu, J.; Streets, D.G. Spatiotemporal dynamics of nitrogen dioxide pollution and urban development: Satellite observations over China, 2005–2016. *Resour. Conserv. Recycl.* **2019**, *142*, 59–68. [CrossRef]
5. Xu, Y.; Wang, D.; Wu, Z. Spatio-temporal Variations of Tropospheric NO₂ over China in 1996–2010 based on Remote Sensing Data. *Remote Sens. Technol. Appl.* **2013**, *28*, 898–903. (In Chinese)
6. Richter, A.; Burrows, J.P. Tropospheric NO₂ from GOME measurements. *Adv. Space Res.* **2002**, *29*, 1673–1683. [CrossRef]
7. Pitters, A.; Bramstedt, K.; Lambert, J.-C.; Kirchhoff, B.J.A.C. Overview of SCIAMACHY validation: 2002–2004. *Atmos. Chem. Phys.* **2006**, *6*, 127–148. [CrossRef]
8. Boersma, K.F.; Jacob, D.J.; Eskes, H.J.; Pinder, R.W.; Wang, J.; Van Der A, R.J. Intercomparison of SCIAMACHY and OMI tropospheric NO₂ columns: Observing the diurnal evolution of chemistry and emissions from space. *J. Geophys. Res.* **2008**, *113*, D16S26.
9. Van Geffen, J.; Eskes, H.; Compernelle, S.; Pinardi, G.; Verhoelst, T.; Lambert, J.C.; Sneep, M.; ter Linden, M.; Ludewig, A.; Boersma, K.F.; et al. Sentinel-5P TROPOMI NO₂ retrieval: Impact of version V2.2 improvements and comparisons with OMI and ground-based data. *Atmos. Meas. Tech.* **2022**, *15*, 2037–2060. [CrossRef]
10. Berger, M.; Moreno, J.; Johannessen, J.A.; Levelt, P.F.; Hanssen, R.F. ESA's sentinel missions in support of Earth system science. *Remote Sens. Environ.* **2012**, *120*, 84–90. [CrossRef]
11. Guanter, L.; Aben, I.; Tol, P.; Krijger, J.M.; Hollstein, A.; Köhler, P.; Damm, A.; Joiner, J.; Frankenberg, C.; Landgraf, J. Potential of the TROPospheric Monitoring Instrument (TROPOMI) onboard the Sentinel-5 Precursor for the monitoring of terrestrial chlorophyll fluorescence. *Atmos. Meas. Tech.* **2015**, *8*, 1337–1352. [CrossRef]
12. Liu, H.F.C.; Huang, J.; Zhu, X.; Zhou, Y.; Wang, Z.; Zhang, Q. The spatial-temporal characteristics and influencing factors of air pollution in Beijing-Tianjin-Hebei urban agglomeration. *Acta Ecol. Sin.* **2018**, *73*, 177–191. (In Chinese)
13. Zheng, Z.; Yang, Z.; Wu, Z.; Marinello, F. Spatial Variation of NO₂ and Its Impact Factors in China: An Application of Sentinel-5P Products. *Remote Sens.* **2019**, *11*, 1939. [CrossRef]
14. Judd, L.M.; Al-Saadi, J.A.; Szykman, J.J.; Valin, L.C.; Janz, S.J.; Kowalewski, M.G.; Eskes, H.J.; Veefkind, J.P.; Cede, A.; Mueller, M.; et al. Evaluating Sentinel-5P TROPOMI tropospheric NO₂ column densities with airborne and Pandora spectrometers near New York City and Long Island Sound. *Atmos. Meas. Tech.* **2020**, *13*, 6113–6140. [CrossRef] [PubMed]
15. Ghasempour, F.; Sekertekin, A.; Kutoglu, S.H. Google Earth Engine based spatio-temporal analysis of air pollutants before and during the first wave COVID-19 outbreak over Turkey via remote sensing. *J. Clean. Prod.* **2021**, *319*, 128599. [CrossRef] [PubMed]
16. Schneider, P.; Hamer, P.D.; Kylling, A.; Shetty, S.; Stebel, K. Spatiotemporal Patterns in Data Availability of the Sentinel-5P NO₂ Product over Urban Areas in Norway. *Remote Sens.* **2021**, *13*, 2095. [CrossRef]
17. Zheng, Z.W.Z.; Chen, Y.; Yang, Z.; Marinello, F. Analysis of temporal and spatial variation characteristics of NO₂ pollutants in Guangdong-Hong Kong-Macao Greater Bay Area based on Sentinel-5P satellite data. *China Environ. Sci.* **2021**, *41*, 63–72.
18. Wang, J.Y.J.; Guo, H.; Wang, X.; Liu, B.; Li, Y. Spatio-temporal analysis of NO₂ column density in Beijing-Tianjin-Hebei region based on TROPOMI. *Environ. Sci. Technol.* **2022**, *45*, 21–31. (In Chinese)
19. He, J.; Hu, S. Ecological efficiency and its determining factors in an urban agglomeration in China: The Chengdu-Chongqing urban agglomeration. *Urban Clim.* **2022**, *41*, 101071. [CrossRef]
20. Lei, Y.; Zhao, J. Development track and new-type urbanization approaches in major conurbation in China. *Acta Sci. Nat. Univ. Sunyatseni* **2016**, *55*, 141–150. (In Chinese)
21. The Central Committee of the Communist Party of China and the State Council issued the “Outline of the Yangtze River Delta Regional Integrated Development Plan”. Available online: <http://www.mofcom.gov.cn/article/b/g/202001/20200102931567.shtml> (accessed on 18 August 2022).
22. Xiao, Z.; Xie, X.; Chen, Y.; Liu, S.; Lin, X.; Chen, G. Temporal and spatial characteristics and influencing factors of NO₂ pollution over Guangdong-Hong Kong-Macao Greater Bay Area. *China Environ. Sci.* **2020**, *40*, 2010–2017.
23. Li, R.; Wang, Z.; Cui, L.; Fu, H.; Zhang, L.; Kong, L.; Chen, W.; Chen, J. Air pollution characteristics in China during 2015–2016: Spatiotemporal variations and key meteorological factors. *Sci. Total Environ.* **2019**, *648*, 902–915. [CrossRef] [PubMed]
24. Wang, J.X.C. Geodetector: Principle and prospective. *Acta Ecol. Sin.* **2017**, *72*, 116–134. (In Chinese)
25. Han, K.M. Temporal Analysis of OMI-Observed Tropospheric NO₂ Columns over East Asia during 2006–2015. *Atmosphere* **2019**, *10*, 658. [CrossRef]

26. Boersma, K.F.; Eskes, H.J.; Dirksen, R.J.; van der A, R.J.; Veefkind, J.P.; Stammes, P.; Huijnen, V.; Kleipool, Q.L.; Sneep, M.; Claas, J.; et al. An improved tropospheric NO₂ column retrieval algorithm for the Ozone Monitoring Instrument. *Atmos. Meas. Tech.* **2011**, *4*, 1905–1928. [CrossRef]
27. Boersma, K.F.; Eskes, H.J.; Veefkind, J.P.; Brinksma, E.J.; van der A, R.J.; Sneep, M.; van den Oord, G.H.J.; Levelt, P.F.; Stammes, P.; Gleason, J.F.; et al. Near-real time retrieval of tropospheric NO₂ from OMI. *Atmos. Chem. Phys.* **2007**, *7*, 2103–2118. [CrossRef]
28. United States Geological Survey (USGS). Available online: <https://lpdaac.usgs.gov/node/838> (accessed on 19 August 2022).
29. Farr, T.G.; Rosen, P.A.; Caro, E.; Crippen, R.; Duren, R.; Hensley, S.; Kobrick, M.; Paller, M.; Rodriguez, E.; Roth, L.; et al. The Shuttle Radar Topography Mission. *Rev. Geophys.* **2007**, *45*, RG2004. [CrossRef]
30. Li, H.; Zhang, J.; Wen, B.; Huang, S.; Gao, S.; Li, H.; Zhao, Z.; Zhang, Y.; Fu, G.; Bai, J.; et al. Spatial-Temporal Distribution and Variation of NO₂ and Its Sources and Chemical Sinks in Shanxi Province, China. *Atmosphere* **2022**, *13*, 1096. [CrossRef]
31. He, Y.; Lin, K.; Liao, N.; Chen, Z.; Rao, J. Exploring the spatial effects and influencing factors of PM_{2.5} concentration in the Yangtze River Delta Urban Agglomerations of China. *Atmos. Environ.* **2022**, *268*, 118805. [CrossRef]
32. Sokal, R.R.; Oden, N.L. Spatial autocorrelation in biology: 1. *Methodology* **1978**, *10*, 199–228.
33. Moran, P.A.P. The interpretation of statistical maps. *J. R. Stat. Soc. Ser. B (Methodol.)* **1948**, *10*, 243–251. [CrossRef]
34. Anselin, L. Local indicators of spatial association-LISA. *Geogr. Anal.* **2010**, *27*, 93–115. [CrossRef]
35. Wang, A.; Kang, P.; Zhang, Y.; Zeng, S.; Zhang, X.; Shi, J.; Liu, Z.; Xiang, Z.; Wang, K.; Zhang, S.; et al. Spatial differentiation and driving factors of aerosol optical depth in Sichuan Basin from 2003 to 2018. *China Environ. Sci.* **2022**, *42*, 528–538. (In Chinese)
36. Zhang, Y.; Wang, L.; Tang, Z.; Zhang, K.; Wang, T. Spatial effects of urban expansion on air pollution and eco-efficiency: Evidence from multisource remote sensing and statistical data in China. *J. Clean. Prod.* **2022**, *367*, 132973. [CrossRef]
37. Zhou, L.; Wu, J.; Jia, R.; Liang, N.; Zhang, F.; Ni, Y.; Liu, M. Investigation of temporal-spatial characteristics and underlying risk factors of PM_{2.5} pollution in Beijing-Tianjin-Hebei Area. *Res. Environ. Sci.* **2016**, *29*, 483–493.
38. Liu, X.; Yi, G.; Zhou, X.; Zhang, T.; Lan, Y.; Yu, D.; Wen, B.; Hu, J. Atmospheric NO₂ Distribution Characteristics and Influencing Factors in Yangtze River Economic Belt: Analysis of the NO₂ Product of TROPOMI/Sentinel-5P. *Atmosphere* **2021**, *12*, 1142. [CrossRef]
39. Liu, Y.; Xie, Y. Remote sensing monitoring of NO₂ concentration based on Sentinel-5P in China. *China Environ. Sci.* **2022**, *42*, 1–16. (In Chinese)
40. Yu, L.; He, W.; Li, N.; Lu, J.; Xie, S.; Huang, L. Analysis of temporal and spatial variation characteristics and influencing factors of NO₂ pollutants in Guangxi. *Environ. Sci. Technol.* **2021**, *44*, 1–9. (In Chinese)
41. Zhou, C.; Wang, Q.; Li, Q.; Liu, S.; Chen, H.; Ma, P.; Wang, Z.; Tan, C. Spatio-temporal change and influencing factors of tropospheric NO₂ column density of Yangtze River Delta in the decade. *China Environ. Sci.* **2016**, *36*, 1921–1930. (In Chinese)
42. Zhao, J.; Cai, K.; Li, S.; Zheng, F.; Liu, Y. Spatiotemporal analysis on the impact of COVID-19 pandemic on NO₂ emission in China. *China Environ. Sci.* **2021**, *41*, 56–62.
43. Li, L.; Li, Q.; Huang, L.; Wang, Q.; Zhu, A.; Xu, J.; Liu, Z.; Li, H.; Shi, L.; Li, R.; et al. Air quality changes during the COVID-19 lockdown over the Yangtze River Delta Region: An insight into the impact of human activity pattern changes on air pollution variation. *Sci. Total Environ.* **2020**, *732*, 139282. [CrossRef] [PubMed]
44. Department of Ecology and Environment of Jiangsu Province. Available online: http://sthjt.jiangsu.gov.cn/art/2020/12/11/art_84025_10211876.html (accessed on 19 August 2022).
45. Falocchi, M.; Zardi, D.; Giovannini, L. Meteorological normalization of NO₂ concentrations in the Province of Bolzano (Italian Alps). *Atmos. Environ.* **2021**, *246*, 118048. [CrossRef]
46. Zhang, Q.; Geng, G.; Wang, S.; Richter, A.; He, K. Satellite remote sensing of changes in NO_x emissions over China during 1996–2010. *Chin. Sci. Bull.* **2012**, *57*, 2857–2864. [CrossRef]
47. Zheng, F.; Yu, T.; Cheng, T.; Gu, X.; Guo, H. Intercomparison of tropospheric nitrogen dioxide retrieved from Ozone Monitoring Instrument over China. *Atmos. Pollut. Res.* **2014**, *5*, 686–695. [CrossRef]
48. Hou, Y.; Wang, L.; Zhou, Y.; Wang, S.; Du, C.; Wang, F. Spatiotemporal variations of tropospheric column nitrogen dioxide over Jing-Jin-Ji during the past decade. *Int. J. Remote Sens.* **2019**, *40*, 15–30. [CrossRef]
49. Shen, Y.; Jiang, F.; Feng, S.; Zheng, Y.; Cai, Z.; Lyu, X. Impact of weather and emission changes on NO₂ concentrations in China during 2014–2019. *Environ. Pollut.* **2021**, *269*, 116163. [CrossRef]
50. Conroy, B.M.; Hamylton, S.M.; Kumbier, K.; Kelleway, J.J. Assessing the structure of coastal forested wetland using field and remote sensing data. *Estuar. Coast. Shelf Sci.* **2022**, *271*, 107861. [CrossRef]
51. Du, Y.; Xie, Z. Global financial crisis making a V-shaped fluctuation in NO₂ pollution over the Yangtze River Delta. *J. Meteorol. Res.* **2017**, *31*, 438–447. [CrossRef]
52. Zheng, L.; Jiang, L.; Chen, S. The economic and social development of one city and three provinces in the Yangtze River Delta will be stable and improving in 2020. *Zhejiang Econ.* **2021**, *38*, 37–40. (In Chinese)
53. Liu, S.; Li, H.; Kun, W.; Zhang, Z.; Wu, H. How Do Transportation Influencing Factors Affect Air Pollutants from Vehicles in China? Evidence from Threshold Effect. *Sustainability* **2022**, *14*, 9402. [CrossRef]
54. Lu, H.; Yao, A.; Yao, C.; Chen, C.; Wang, B. An investigation on the characteristics of and influence factors for NO₂ formation in diesel/methanol dual fuel engine. *Fuel* **2019**, *235*, 617–626. [CrossRef]
55. Liu, F.; Xing, C.; Su, P.; Luo, Y.; Zhao, T.; Xue, J.; Zhang, G.; Qin, S.; Song, Y.; Bu, N. Source analysis of the tropospheric NO₂ based on MAX-DOAS measurements in northeastern China. *Environ. Pollut.* **2022**, *306*, 119424. [CrossRef] [PubMed]